

TrES-2b

R-0123

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_170512 · created 2026-07-16T05:23:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457885.91107 +/- 0.00051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.04
Transit depth (Rp/Rs)^2	2.14 +/- 1.18 [%]
Orbital Inclination (inc)	84.44 +/- 2.42
Airmass coefficient 1 (a1)	1.697 +/- 0.018
Airmass coefficient 2 (a2)	-0.5 +/- 0.22
Scatter in the residuals of the lightcurve fit is	1.04 %
Best Comparison Star	None
Optimal Aperture	4.19
Optimal Annulus	10.61
Transit Duration (day)	0.041 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.146 ± 0.04 vs 0.1254 (deviation 0.48σ)
Duration fit vs published	0.041 d vs 0.0746 d (deviation 1.76σ)
Mid-time O–C	0.4 min
Depth SNR	3.4
Residual scatter	1.04 %

Light curve

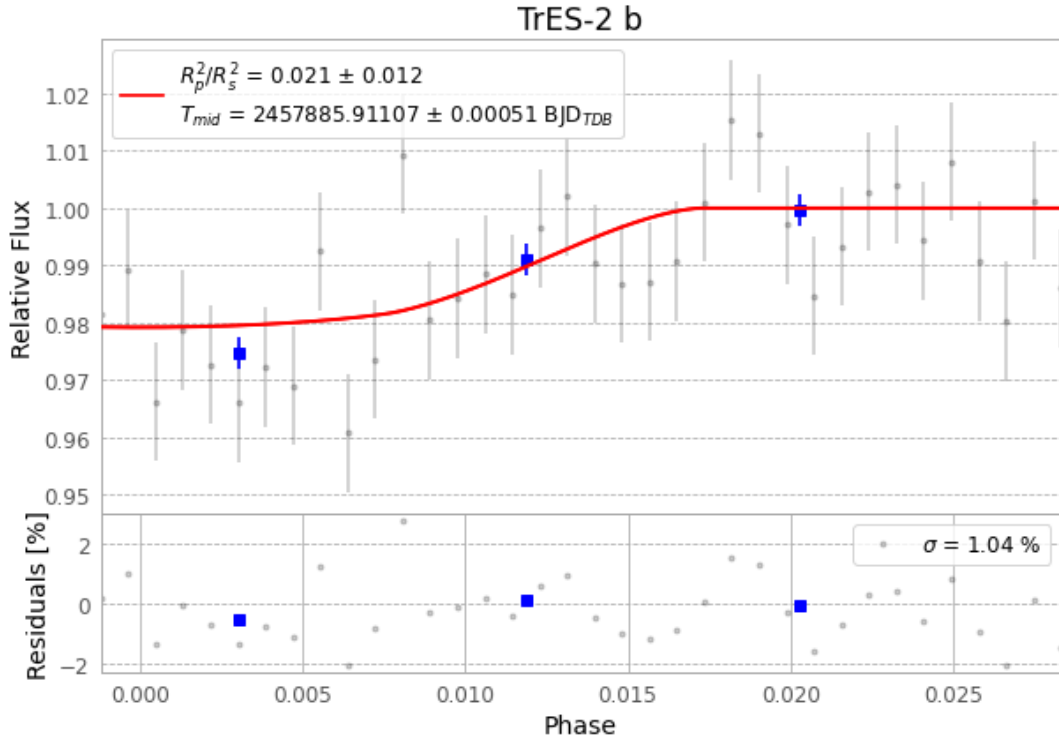


Figure 1 — FinalLightCurve_TrES-2 b_2017-05-12.png (SHA-256 f8c2169182f1b381...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [496, 273], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 78093eb1e37d9647...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

36 FITS frames (24 MB), TRES-2170512094512.FITS → TRES-2170512113012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-170512.log (544a1b0cf6ba...), FinalLightCurve_TrES-2 b_2017-05-12.png (f8c2169182f1...), FinalLightCurve_TrES-2 b_2017-05-12.csv (b4c191163184...)

Compute resources

Wall time 170.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:23Z.